Nicolò **Tampellini**

Postdoctoral Associate · Organic Chemistry

Cambridge, MA, USA

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Education

Ph.D. / Yale University *New Haven, CT, USA*

Doctoral Student *Sep 2021 - Feb 2026*

- **Advisor:** Prof. Scott J. Miller
- **Dissertation Title:** "On Atomistic Modeling and Asymmetric Catalysis: Stereocontrolled Synthesis by Design"

B.Sc. / Alma Mater Studiorum - Università di Bologna *Bologna, IT*

Bachelor of Science Degree *Sep 2017 - Jul 2020*

- **Final Grade:** 110 *cum laude* / 110 (Italian GPA 29.43 / 30)
- **Thesis advisor:** Prof. Paolo Righi - Co-advisor: Prof. Giorgio Bencivenni
- **Dissertation Title:** "Computational Insights into the Enantioselective Axial Desymmetrization of Maleimides"

Experience

Postdoctoral Associate / MIT *Cambridge, MA, USA*

MIT - Advisor: Prof. Alexander T. Radosevich *May 2026 - present*

- Serving as **Student Seminar Committee Member** since June 2026. My responsibilities include the invitation, organization and hosting of seminar speakers in organic chemistry as part of MIT's student-hosted seminar series.

Teaching Fellow - Org. Chem. Lab. (UG) and Advanced Organic Chemistry (Graduate) *New Haven, CT, USA*

Yale University - Advisor: Prof. Scott J. Miller *Sep 2021 - Dec 2023*

- Served as a **Laboratory Teaching Fellow** (TF) for the Organic Chemistry Laboratory class for two semesters to groups of ~30 students per semester. My responsibility included teaching and demonstrating laboratory procedures, grading weekly assignments and exams and offering personalized guidance to individual students.
- Served as a **Lecture Teaching Fellow** (TF) for Prof.s William L. Jorgensen and Scott J. Miller's class on graduate level physical organic chemistry, focusing on conformational energetics, organic thermochemistry, kinetics, thermodynamics, orbital symmetry and stereochemistry. My responsibilities included hosting weekly exercise sessions, clarifying complex concepts from the class, grading weekly assignments and exams, and offering personalized guidance to individual students.

Host - Computational Chemistry Lecture Series *New Haven, CT, USA*

Yale University - Advisor: Prof. Scott J. Miller *Oct 2025 - Feb 2026*

- Hosted a weekly, open-door Computational Chemistry Lecture Series about the theoretical and practical aspects of running computational atomistic modeling calculations with hands-on tutorials and personalized application to individual research projects of the attendees. My responsibilities included the preparation of each session's materials, the coding of user-specific functionality within FIRECODE (see Personal Projects below) and the exposition of recent publications in the computational organic chemistry literature. The lectures were consistently attended by the majority of the lab members of my research group (>10 people).

Awards

2026	Richard Wolfgang Prize (Best Doctoral Thesis in Chemistry), Yale University - New Haven, CT - Link
2024	Kenneth B. Wiberg Research Fellowship (Competitive, one-year), Yale University - New Haven, CT - Link
2017	National Finalist, Italian Olympiads of Chemistry - Rome, IT - Link
2016	National Finalist, Italian Olympiads of Chemistry - Frascati, IT - Link

Publications

9. A Practical Catalytic Enantioselective Synthesis of Fully Heteroatom-Substituted P(V) Species
Tampellini, N.‡; Guerrero, M.‡; Miller, S.* *in preparation as of August 2026*

‡ = equal contribution

8. FIRECODE: A Computational Chemistry Workflow Driver and Modular Hub
Tampellini, N.*, *JOSS* **2026**, 11(124), 10698 - Link - Public Review

7. A Bifunctional Aminoxyl–Bipyridine Peptide Catalyst for the Atroposelective Copper-Catalyzed Aerobic Oxidation of Biaryl Diols
Marvich, H.; Langner, O.; **Tampellini, N.**; Miller, S.* *J. Am. Chem. Soc.*, **2026**, 148, 23032-23041 - Link

• Highlighted by The American Peptide Society - Link

6. Peptide-Catalyzed Asymmetric Amination of Sulfenamides Enabled by DFT-Guided Catalyst Optimization
Tampellini, N.; Choi, E. S.; Miller, S.* *J. Am. Chem. Soc.*, **2025**, *147*, 41122-41129 - Link

- Highlighted by The American Peptide Society - Link
- Highlighted by Thieme Synfacts - Link
- Highlighted by Chemistry Europe - Link

5. Exp.tal Lineage and Computational Analysis of a General Aminoxyl-based Oxidation Catalyst: Generality from Substrate-Specific Interactions
Rozema, S.‡; **Tampellini, N.‡**; Rein, J.; Sigman, M.; Lin, S.; Miller, S.* *ACS Catalysis* **2025**, 15, *20*, 17548–17557 - Link

‡ = equal contribution, listed in alphabetical order

4. Enantiocontrolled Cyclization to Form Chiral 7- and 8-Membered Rings Unified by the Same Catalyst Operating with Different Mechanisms
Tampellini, N., Mercado, B., and Miller, S.* *J. Am. Chem. Soc.* **2025**, 147, 5, 4624–4630 - Link

- Highlighted with a Research Spotlight on Synthesis Workshop - Link

3. Catalytic Enantioselective Sulfoxidation of Functionalized Thioethers Mediated by Aspartic Acid-Containing Peptides
Huth, S., **Tampellini, N.**, Guerrero, M. and Miller, S.* *Org. Lett.* **2024**, 26, 32, 6872–6877 - Link

2. Scaffold-Oriented Asymmetric Catalysis: Conformational Modulation of Transition State Multivalency during a Catalyst-Controlled Assembly of a Pharmaceutically Relevant Atropisomer
Tampellini, N.; Mercado, B. and Miller, S.* *Chem. Eur. J.* **2024**, e202401109 - Link

1. Computational Investigation on the Origin of Atroposelectivity for the Cinchona Alkaloid Primary Amine-Catalyzed Vinylogous Desymmetrization of N-(2-t-Butylphenyl)maleimides
Tampellini, N.*; Righi, P. and Bencivenni, G.* *J. Org. Chem.* **2021**, 86, 17, 11782–11793 - Link

Personal Projects

FIRECODE - Filtering Refiner and Embedder for Conformationally Dense Ensembles
2021-present

- Open-source computational chemistry workflow driver and modular hub. Enables the automation of molecular modeling tasks encompassing large and complex conformational spaces, including ground and transition state ensemble generation, optimization, structural manipulation and thermochemistry. Interfaces with external calculators like XTB, TBLITE, ORCA, and Neural Network potentials (AIMNET2, UMA). Repository - Paper

PRISM - Pruning Interface for Similar Molecules (with Rowan Scientific Corporation)
2025-present

- Open-source program for similarity pruning of molecular ensembles. Modular production-ready code to remove chemically identical, rotameric and mirror image molecular duplicates from conformational ensembles. Adopted by Rowan as part of the company's pipeline. A fast conformational search algorithm based on PRISM's fast similarity pruning named *openconf* also demonstrates the impact of the software. Repository - Guest Article - openconf Article

Conferences and Invited Talks

Stereochemistry Gordon Research Seminar 2026 / Invited Speaker
Newport, RI, USA
SALVE REGINA UNIVERSITY *July 2026*

- Presented my research work on "A Practical Catalytic Enantioselective Synthesis of Fully Heteroatom-Substituted P(V) Species".

Rowan Seminar Series / Invited Speaker
Boston, MA, USA
ROWAN SCIENTIFIC CORPORATION - HOST: DR. JONATHON VANDEZANDE *January 2026*

- Presented my research work on "Atomistic Modeling to Understand and Design Flexible Asymmetric Catalysts - Link

National Organic Symposium 2025 / Poster Presenter
Troy, NY, USA
RENSSELAER POLYTECHNIC INSTITUTE *July 2025*

- Presented a poster about my research work on the "Stereocontrolled Cyclization of Inherently Chiral Medium-Sized Rings".

Pfizer Research Spotlight / Poster Presenter
Cambridge, MA, USA
PFIZER KENDALL SQUARE SITE *June 2025*

- Presented my research work on the "Stereocontrolled Cyclization of Inherently Chiral Medium-Sized Rings" to an audience of Pfizer professionals and selected peers.

Indena Innovation Symposium 2024 / Invited Speaker
Settala, IT
INDENA S.P.A. - HOST: ALESSANDRO BRUSA *Dec. 2024*

- Presented my research work on the "Stereocontrolled Cyclization of Inherently Chiral Medium-Sized Rings" to an audience of professionals working on the Research and Development branch of the company.

Yale Chemistry Symposium 2024 / Invited Speaker
New Haven, CT, USA
YALE UNIVERSITY - HOST: PROF. SCOTT J. MILLER *Aug. 2024*

- Presented my research work to the department on the "Stereocontrolled Cyclization of Inherently Chiral Medium-Sized Rings"

UTD Research Talk 2024 / Invited Speaker *Richardson, TX, USA*

UNIVERSITY OF TEXAS AT DALLAS - HOST: PROF. FILIPPO ROMITI *June 2024*

- Presented my research work titled ”Atroposelective Synthesis of New Chiral Scaffolds with Flexible, Bifunctional Superbases”

Organic Chemistry Colloquium 2023 / Invited Speaker *Bologna, IT*

UNIVERSITY OF BOLOGNA - HOST: PROF. PAOLO RIGHI *Dec. 2023*

- Presented my research work on ”Catalyst Conformational Modulation - Enabling the Atroposelective Cyclization of Quinazolinediones with Flexible Bifunctional Superbases”